Program Summary - Ridership_Regression.sas

Execution Environment

Author:	u64420076
File:	/home/u64420076/BAN000_Personal/Ridership_Regression.sas
SAS Platform:	Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host:	ODAWS01-USW2.ODA.SAS.COM
SAS Version:	9.04.01M8P02222023
SAS Locale:	en_US
Submission Time:	4/15/2026, 1:40:32 PM
Browser Host:	POOL-173-35-226-51.CPE.NET.CABLE.ROGERS.COM
User Agent:	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36
Application Server:	ODAMID00-USW2.ODA.SAS.COM

Code: Ridership_Regression.sas

```
/* Importr Data */

FILENAME REFFILE '/home/u64420076/BAN000_Personal/main_dataset_modeling.csv';

PROC IMPORT DATAFILE=REFFILE
    DBMS=CSV
    OUT=WORK.TRANSIT;
    GETNAMES=YES;
RUN;

/* Baseline Time Structure */


proc genmod data=transit;
    class month (ref='12') covid_period (ref='Pre-COVID') / param=ref;
    model ridership =
        month
        covid_period
        / dist=normal link=identity type3;
    title "Model 1: Baseline Time Structure";
run;

/* Lagged Ridership Baseline */
proc genmod data=transit;
    class month (ref='12') covid_period (ref='Pre-COVID') / param=ref;
    model ridership =
        month
        covid_period
        ridership_lag1
        ridership_lag12
        / dist=normal link=identity type3;
    title "Model 2: Lagged Ridership Baseline";
run;

/* Full Model with External Variables */
proc genmod data=transit;
    class month (ref='12') covid_period (ref='Pre-COVID') / param=ref;
    model ridership =
        month
        covid_period
        ridership_lag1
        ridership_lag12
        gas_price_lag1
        unemployment_rate_lag1
        avg_temp
        total_precip_mm
        total_snow_cm
        / dist=normal link=identity type3;
    title "Model 3: Full Model with External Variables";
run;
```

Log: Ridership_Regression.sas

Notes (12)

```
1          OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69         /* Importr Data */
70
71         FILENAME REFFILE '/home/u64420076/BAN000_Personal/main_dataset_modeling.csv';
72
73         PROC IMPORT DATAFILE=REFFILE
74         DBMS=CSV
75         OUT=WORK.TRANSIT;
76         GETNAMES=YES;
77         RUN;

NOTE: Import cancelled.  Output dataset WORK.TRANSIT already exists.  Specify REPLACE option to overwrite it.
NOTE: The SAS System stopped processing this step because of errors.
NOTE: PROCEDURE IMPORT used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              197.09k
      OS Memory           26024.00k
      Timestamp           04/15/2026 05:40:32 PM
      Step Count                        39  Switch Count  0
      Page Faults                       0
      Page Reclaims                     14
      Page Swaps                        0
      Voluntary Context Switches        0
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           0

78
79         /* Baseline Time Structure */
80
81


82         proc genmod data=transit;
83             class month (ref='12') covid_period (ref='Pre-COVID') / param=ref;
84             model ridership =
85                 month
86                 covid_period
87                 / dist=normal link=identity type3;
88             title "Model 1: Baseline Time Structure";
89         run;

NOTE: Algorithm converged.
NOTE: The scale parameter was estimated by maximum likelihood.
NOTE: PROCEDURE GENMOD used (Total process time):
      real time           0.07 seconds
      user cpu time       0.07 seconds
      system cpu time     0.00 seconds
      memory              2375.46k
      OS Memory           26800.00k
      Timestamp           04/15/2026 05:40:32 PM
      Step Count                        40  Switch Count  1
      Page Faults                       0
      Page Reclaims                     327
      Page Swaps                        0
      Voluntary Context Switches        9
      Involuntary Context Switches      3
      Block Input Operations            0
      Block Output Operations           48


90
91         /* Lagged Ridership Baseline */
92         proc genmod data=transit;
93             class month (ref='12') covid_period (ref='Pre-COVID') / param=ref;
94             model ridership =
95                 month
96                 covid_period
97                 ridership_lag1
98                 ridership_lag12
99                 / dist=normal link=identity type3;
100            title "Model 2: Lagged Ridership Baseline";
101        run;

NOTE: Algorithm converged.
NOTE: The scale parameter was estimated by maximum likelihood.
NOTE: PROCEDURE GENMOD used (Total process time):
      real time           0.07 seconds
      user cpu time       0.07 seconds
      system cpu time     0.00 seconds
      memory              1384.50k
      OS Memory           27056.00k
      Timestamp           04/15/2026 05:40:32 PM
      Step Count                        41  Switch Count  1
      Page Faults                       0
      Page Reclaims                     139
```

```
Page Swaps                        0
Voluntary Context Switches        12
Involuntary Context Switches      7
Block Input Operations            0
Block Output Operations           40


102
103        /* Full Model with External Variables */
104        proc genmod data=transit;
105            class month (ref='12') covid_period (ref='Pre-COVID') / param=ref;
106            model ridership =
107                month
108                covid_period
109                ridership_lag1
110                ridership_lag12
111                gas_price_lag1
112                unemployment_rate_lag1
113                avg_temp
114                total_precip_mm
115                total_snow_cm
116                / dist=normal link=identity type3;
117            title "Model 3: Full Model with External Variables";
118        run;

NOTE: Algorithm converged.
NOTE: The scale parameter was estimated by maximum likelihood.
NOTE: PROCEDURE GENMOD used (Total process time):
      real time           0.08 seconds
      user cpu time       0.07 seconds
      system cpu time     0.00 seconds
      memory              1413.43k
      OS Memory           27056.00k
      Timestamp           04/15/2026 05:40:32 PM
      Step Count                        42  Switch Count  1
      Page Faults                       0
      Page Reclaims                     132
      Page Swaps                        0
      Voluntary Context Switches        11
      Involuntary Context Switches      2
      Block Input Operations            0
      Block Output Operations           56


119
120
121
122        OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
132
```

Results: Ridership_Regression.sas

Model 1: Baseline Time Structure

The GENMOD Procedure

Model Information	
Data Set	WORK.TRANSIT
Distribution	Normal
Link Function	Identity
Dependent Variable	ridership

Number of Observations Read	120
Number of Observations Used	120

Class Level Information												
Class	Value	Design Variables										
month	1	1	0	0	0	0	0	0	0	0	0	0
	2	0	1	0	0	0	0	0	0	0	0	0
	3	0	0	1	0	0	0	0	0	0	0	0
	4	0	0	0	1	0	0	0	0	0	0	0
	5	0	0	0	0	1	0	0	0	0	0	0
	6	0	0	0	0	0	1	0	0	0	0	0
	7	0	0	0	0	0	0	1	0	0	0	0
	8	0	0	0	0	0	0	0	1	0	0	0
	9	0	0	0	0	0	0	0	0	1	0	0
	10	0	0	0	0	0	0	0	0	0	1	0
	11	0	0	0	0	0	0	0	0	0	0	1
	12	0	0	0	0	0	0	0	0	0	0	0
covid_period	COVID/Rec	1	0									
	Post-COVI	0	1									
	Pre-COVID	0	0									

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Deviance	106	2.4199891E12	22830085486
Scaled Deviance	106	120.0000	1.1321
Pearson Chi-Square	106	2.4199891E12	22830085486
Scaled Pearson X2	106	120.0000	1.1321
Log Likelihood		-1593.9102	
Full Log Likelihood		-1593.9102	
AIC (smaller is better)		3217.8203	
AICC (smaller is better)		3222.4357	
BIC (smaller is better)		3259.6327	

Algorithm converged.

Analysis Of Maximum Likelihood Parameter Estimates								
Parameter		DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept		1	1635444	47614.34	1542121	1728766	1179.77	<.0001
month	1	1	-33788.9	63587.44	-158418	90840.17	0.28	0.5952
month	2	1	13911.08	63587.44	-110718	138540.2	0.05	0.8268
month	3	1	84000.00	63508.39	-40474.1	208474.1	1.75	0.1859
month	4	1	-2100.00	63508.39	-126574	122374.1	0.00	0.9736
month	5	1	14600.00	63508.39	-109874	139074.1	0.05	0.8182
month	6	1	35700.00	63508.39	-88774.1	160174.1	0.32	0.5740
month	7	1	20800.00	63508.39	-103674	145274.1	0.11	0.7433
month	8	1	3200.000	63508.39	-121274	127674.1	0.00	0.9598
month	9	1	160400.0	63508.39	35925.85	284874.1	6.38	0.0115
month	10	1	152800.0	63508.39	28325.85	277274.1	5.79	0.0161
month	11	1	154900.0	63508.39	30425.85	279374.1	5.95	0.0147
covid_period	COVID/Rec	1	-973889	31697.64	-1036015	-911763	943.99	<.0001
covid_period	Post-COVI	1	-411257	31062.35	-472138	-350376	175.29	<.0001
Scale		1	142009.1	9166.646	125132.8	161161.4		

Note: The scale parameter was estimated by maximum likelihood.

LR Statistics For Type 3 Analysis			
Source	DF	Chi-Square	Pr > ChiSq
month	11	23.72	0.0140
covid_period	2	261.92	<.0001

Model 2: Lagged Ridership Baseline

The GENMOD Procedure

Model Information	
Data Set	WORK.TRANSIT
Distribution	Normal
Link Function	Identity
Dependent Variable	ridership

Number of Observations Read	120
Number of Observations Used	120

Class Level Information												
Class	Value	Design Variables										
month	1	1	0	0	0	0	0	0	0	0	0	0
	2	0	1	0	0	0	0	0	0	0	0	0
	3	0	0	1	0	0	0	0	0	0	0	0
	4	0	0	0	1	0	0	0	0	0	0	0
	5	0	0	0	0	1	0	0	0	0	0	0
	6	0	0	0	0	0	1	0	0	0	0	0
	7	0	0	0	0	0	0	1	0	0	0	0
	8	0	0	0	0	0	0	0	1	0	0	0
	9	0	0	0	0	0	0	0	0	1	0	0
	10	0	0	0	0	0	0	0	0	0	1	0
	11	0	0	0	0	0	0	0	0	0	0	1
	12	0	0	0	0	0	0	0	0	0	0	0
covid_period	COVID/Rec	1	0									
	Post-COVI	0	1									
	Pre-COVID	0	0									

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Deviance	104	890427008538	8561798159.0
Scaled Deviance	104	120.0000	1.1538
Pearson Chi-Square	104	890427008538	8561798159.0
Scaled Pearson X2	104	120.0000	1.1538
Log Likelihood		-1533.9211	
Full Log Likelihood		-1533.9211	
AIC (smaller is better)		3101.8423	
AICC (smaller is better)		3107.8423	

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
BIC (smaller is better)		3149.2296	

Algorithm converged.

Analysis Of Maximum Likelihood Parameter Estimates								
Parameter		DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept		1	800309.5	105980.5	592591.5	1008028	57.02	<.0001
month	1	1	91374.36	39914.29	13143.78	169604.9	5.24	0.0221
month	2	1	133048.0	39599.34	55434.69	210661.2	11.29	0.0008
month	3	1	120374.4	38612.22	44695.80	196052.9	9.72	0.0018
month	4	1	40011.63	38677.05	-35794.0	115817.3	1.07	0.3009
month	5	1	110257.4	39260.09	33309.02	187205.8	7.89	0.0050
month	6	1	123491.0	39107.45	46841.79	200140.2	9.97	0.0016
month	7	1	93509.66	38949.49	17170.06	169849.2	5.76	0.0164
month	8	1	82287.79	39073.42	5705.293	158870.3	4.44	0.0352
month	9	1	272221.7	39303.19	195188.9	349254.6	47.97	<.0001
month	10	1	171423.6	38736.70	95501.06	247346.1	19.58	<.0001
month	11	1	178167.6	38732.21	102253.9	254081.3	21.16	<.0001
covid_period	COVID/Rec	1	-504979	55957.38	-614653	-395304	81.44	<.0001
covid_period	Post-COVI	1	-236009	31834.95	-298405	-173614	54.96	<.0001
ridership_lag1		1	0.5945	0.0486	0.4993	0.6898	149.82	<.0001
ridership_lag12		1	-0.1394	0.0255	-0.1894	-0.0893	29.82	<.0001
Scale		1	86140.73	5560.360	75903.83	97758.24		

Note: The scale parameter was estimated by maximum likelihood.

LR Statistics For Type 3 Analysis			
Source	DF	Chi-Square	Pr > ChiSq
month	11	54.87	<.0001
covid_period	2	62.17	<.0001
ridership_lag1	1	97.23	<.0001
ridership_lag12	1	26.63	<.0001

Model 3: Full Model with External Variables

The GENMOD Procedure

Model Information	
Data Set	WORK.TRANSIT
Distribution	Normal
Link Function	Identity
Dependent Variable	ridership

Number of Observations Read	120
Number of Observations Used	120

Class Level Information											
Class	Value	Design Variables										
month	1	1	0	0	0	0	0	0	0	0	0	0
	2	0	1	0	0	0	0	0	0	0	0	0
	3	0	0	1	0	0	0	0	0	0	0	0
	4	0	0	0	1	0	0	0	0	0	0	0
	5	0	0	0	0	1	0	0	0	0	0	0
	6	0	0	0	0	0	1	0	0	0	0	0
	7	0	0	0	0	0	0	1	0	0	0	0
	8	0	0	0	0	0	0	0	1	0	0	0
	9	0	0	0	0	0	0	0	0	1	0	0
	10	0	0	0	0	0	0	0	0	0	1	0
	11	0	0	0	0	0	0	0	0	0	0	1
	12	0	0	0	0	0	0	0	0	0	0	0
covid_period	COVID/Rec	1	0									
	Post-COVI	0	1									
	Pre-COVID	0	0									

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Deviance	99	634651534889	6410621564.5
Scaled Deviance	99	120.0000	1.2121
Pearson Chi-Square	99	634651534889	6410621564.5
Scaled Pearson X2	99	120.0000	1.2121
Log Likelihood		-1513.6036	
Full Log Likelihood		-1513.6036	
AIC (smaller is better)		3071.2073	
AICC (smaller is better)		3081.6403	
BIC (smaller is better)		3132.5321	

Algorithm converged.

Analysis Of Maximum Likelihood Parameter Estimates								
Parameter		DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept		1	252910.6	145625.9	-32510.8	538332.1	3.02	0.0824
month	1	1	110575.9	36134.52	39753.51	181398.2	9.36	0.0022
month	2	1	128824.2	34934.19	60354.47	197294.0	13.60	0.0002
month	3	1	105698.6	34890.29	37314.87	174082.3	9.18	0.0024
month	4	1	-11664.2	47166.71	-104109	80780.85	0.06	0.8047
month	5	1	-4220.61	70879.29	-143141	134700.2	0.00	0.9525
month	6	1	-47258.8	93203.76	-229935	135417.2	0.26	0.6121
month	7	1	-96699.0	106432.8	-305303	111905.4	0.83	0.3636
month	8	1	-100198	101953.5	-300023	99627.41	0.97	0.3257
month	9	1	108032.9	87637.24	-63732.9	279798.7	1.52	0.2177
month	10	1	72740.99	60024.92	-44905.7	190387.7	1.47	0.2256
month	11	1	125628.3	39260.08	48679.92	202576.6	10.24	0.0014
covid_period	COVID/Rec	1	-626488	52157.07	-728714	-524262	144.28	<.0001
covid_period	Post-COVI	1	-377909	34243.71	-445025	-310792	121.79	<.0001
ridership_lag1		1	0.5260	0.0574	0.4134	0.6385	83.86	<.0001
ridership_lag12		1	-0.0498	0.0301	-0.1088	0.0091	2.75	0.0972
gas_price_lag1		1	3681.719	553.1004	2597.663	4765.776	44.31	<.0001
unemployment_rate_la		1	18975.11	6907.697	5436.274	32513.95	7.55	0.0060
avg_temp		1	5919.512	4527.187	-2953.61	14792.64	1.71	0.1910
total_precip_mm		1	37.1015	223.4643	-400.880	475.0835	0.03	0.8681
total_snow_cm		1	-140.161	870.1469	-1845.62	1565.295	0.03	0.8720
Scale		1	72723.88	4694.306	64081.43	82531.90		

Note: The scale parameter was estimated by maximum likelihood.

LR Statistics For Type 3 Analysis			
Source	DF	Chi-Square	Pr > ChiSq
month	11	57.41	<.0001
covid_period	2	96.32	<.0001
ridership_lag1	1	63.59	<.0001
ridership_lag12	1	2.72	0.0992
gas_price_lag1	1	37.71	<.0001
unemployment_rate_la	1	7.32	0.0068
avg_temp	1	1.70	0.1926
total_precip_mm	1	0.03	0.8681
total_snow_cm	1	0.03	0.8720